

VL-Nav: Neuro-Symbolic Reasoning-based Vision-Language Navigation

Abstract—Navigating unseen, large-scale environments from complex and abstract human instructions requires robots to infer implicit semantics and efficiently explore large task spaces. Existing end-to-end and foundation-model navigation systems often lack either the reasoning needed to decompose complex tasks or the exploration mechanisms needed to avoid aimless travel. We present VL-Nav, a neuro-symbolic (NeSy) vision-language navigation system with two tightly coupled components: (1) a NeSy Task Planner that grounds vision-language-model reasoning in a symbolic 3D scene graph and object-centric image memory for task decomposition and replanning; and (2) a NeSy Exploration System that combines neural semantic cues with geometric and curiosity-based heuristics to gather task-relevant information while reducing repeated travel. On DARPA TIAMAT navigation tasks, VL-Nav achieves an 83.4% success rate (SR) indoors and 75.0% outdoors. Across four unseen physical environments, it reaches 86.3% SR, including a 483-m run. We further demonstrate complex instruction following in a multi-floor environment.

Index Terms—vision-language navigation, neuro-symbolic reasoning, semantic mapping, embodied AI

I. INTRODUCTION

Efficient navigation following complex human instructions in completely unseen environments remains a formidable challenge. Consider a DARPA TIAMAT task [1]: “Today’s weather report indicates rain. Help Rob find an umbrella, an appropriate jacket, and shoes.” The robot must bridge a substantial logical gap: infer that the targets should be waterproof, distinguish a rain jacket and rain boots from visually similar alternatives, sequence multiple goals, and search an unknown map within a time limit. This moves beyond direct command-following vision-language navigation (VLN) [2] toward reasoning-based VLN.

Classical exploration [3], [4], [5] provides geometric coverage but cannot resolve implicit language. End-to-end reinforcement-learning and vision-language-action (VLA) policies [6], [7], [8], [9], [10] learn semantic priors but are data-hungry, computationally expensive, and vulnerable to sim-to-real shift [11]. Foundation-model modular systems such as VLFM [12], SG-Nav [13], and ApexNav [14] offer zero-shot object navigation, but commonly assume an explicitly named target. Their tight coupling between target verification and exploration also limits multi-target reasoning, while an over-reliance on neural relevance can cause repeated travel through already explored regions.

VL-Nav addresses two capabilities jointly: *accurate task reasoning* for abstract, multi-target instructions and *efficient semantic exploration* in large-scale environments. The system contributes:

- a NeSy task planner that decomposes and replans against a unified symbolic memory containing a 3D scene graph and best-view object images;

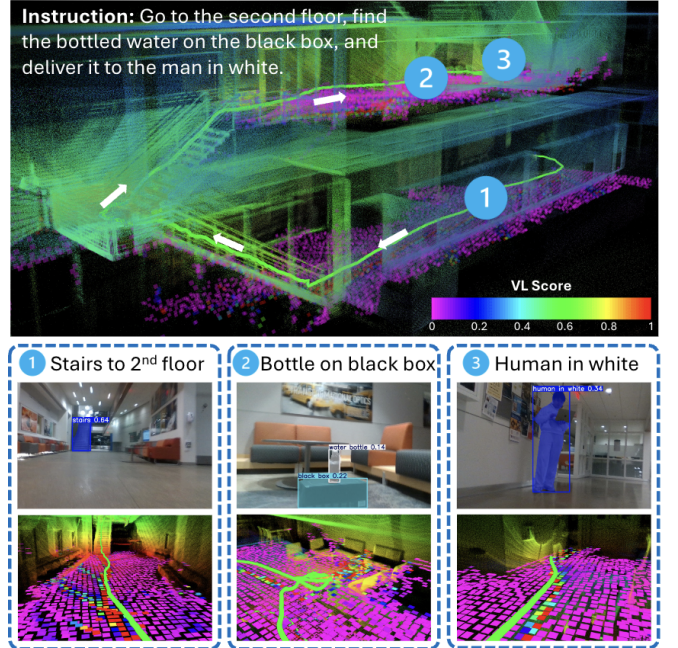


Fig. 1. VL-Nav executes a complex multi-floor instruction. The trajectory and projected value map show task-directed exploration. The system finds stairs, grounds the water bottle using the relation “on the black box,” and verifies the recipient from the attribute “in white.”

- a real-time NeSy exploration policy that combines open-vocabulary semantic evidence with frontiers, travel cost, and information gain; and
- evaluation in four simulated and four physical environments, reaching 79.2% overall simulation SR and 86.3% real-world SR, with long-range and multi-floor demonstrations such as Fig. 1.

II. RELATED WORK

Classical and end-to-end navigation. SLAM-based systems construct geometric maps and localize a robot as the map grows [15], [16], [17]. Frontier heuristics select boundaries between known free and unknown space [5], [18]; they are effective for coverage but do not connect visual semantics to language. End-to-end navigation instead maps observations and instructions to actions using imitation or reinforcement learning [19], [20]. Newer VLA systems unify vision, language, and action in large transformers. Their learned priors are useful, but the training burden, limited interpretability, and distribution shift remain problematic for real robots.

Semantic and foundation-model navigation. Modular methods inject learned perception into classical mapping and planning [21], [22], [23]. Foundation models extend

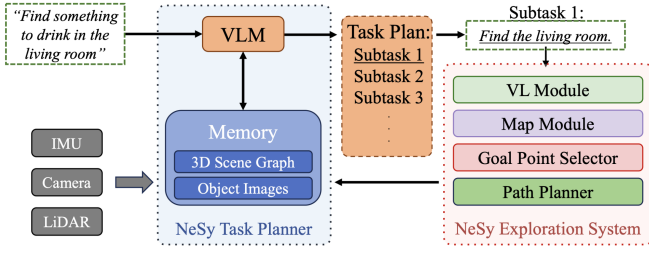


Fig. 2. VL-Nav pipeline. The NeSy Task Planner decomposes an instruction while consulting the unified memory. Its active subtask drives the VL and map modules, goal selection, and collision-free path planning.

this idea with open-vocabulary perception and language reasoning. VLFM [12] projects image-language relevance onto frontiers; SG-Nav [13] reasons over an online 3D scene graph; and ApexNav [14] uses target-centric semantic fusion. VL-Nav is distinguished by reasoning over abstract, multi-target instructions and by separating asynchronous high-level VLM reasoning from a lightweight, continuously running exploration system.

III. METHODOLOGY

As shown in Fig. 2, VL-Nav closes a loop between a deliberative task planner, persistent symbolic memory, and real-time exploration. The planner is invoked when a task arrives or a subtask terminates; perception, mapping, goal selection, and motion planning run continuously.

A. NeSy Task Planner

Unified symbolic memory. Inspired by Hydra [24], the memory contains room and object nodes. Room masks are extracted from the occupancy map using morphological operations and labeled by an LLM from the objects they contain. An object node is connected to a room when its centroid lies within the room mask. Each object stores four attributes: its 3D centroid, highest detection confidence, the observing robot pose, and the corresponding best-view RGB image. The scene graph therefore supports compact spatial and relational queries, while image memory preserves appearance evidence that would be lost in a symbolic label alone.

Task decomposition and replanning. A Qwen3-VL backbone [25] converts the instruction into two atomic subtask types: *explore*, which gathers evidence for a prompt, and *go to*, which approaches and reports a localized target. The model maintains a dynamic target list and replans after each subtask using the updated memory. Target verification is coarse-to-fine. First, the scene graph filters object nodes to the top- k candidates by category, confidence, and spatial relations. Second, the VLM inspects their best-view images and neighboring nodes to select the candidate that best matches the abstract instruction. The stored viewpoint of that node becomes the navigation goal. This grounds each decision in observed, localized evidence and avoids running a large VLM at every motion step.

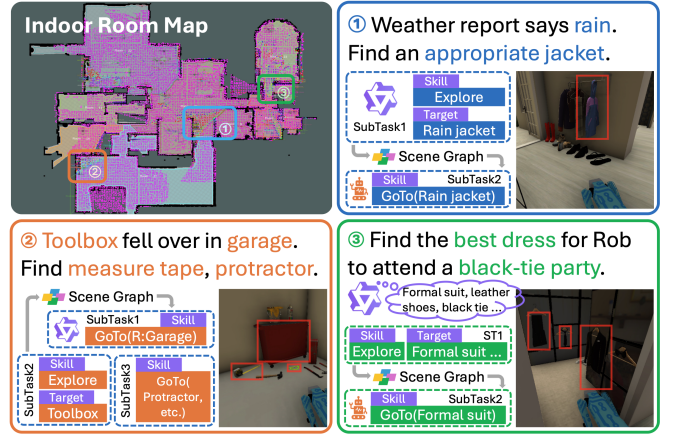


Fig. 3. Qualitative reasoning in the TIAMAT indoor environment: implicit inference (rain $\rightarrow$ rain jacket), hierarchical decomposition (garage $\rightarrow$ tools), and abstract grounding (black-tie party $\rightarrow$ formal attire).

B. NeSy Exploration System

The exploration system receives the active subtask and forms a candidate-goal pool. A free cell is a frontier when at least one neighboring cell is unknown. Breadth-first search clusters frontiers observed in the robot's forward field of view; small clusters yield one centroid and large clusters yield multiple sampled goals. In parallel, open-vocabulary detections above threshold τ_{det} are projected into the map as *instance-based target points* (IBTPs) and voxel-downsampled. IBTPs serve as semantic shortcuts: a tentative observation can be approached for verification instead of being discarded or waiting for complete certainty.

YOLO-World [26] and FastSAM [27] provide pixel-wise, open-vocabulary cues. Suppose the detections define K angular regions $(\mu_k, \sigma_k, \alpha_k)$. For goal direction offset $\Delta\theta$, their accumulated score is

$$S_{\text{VL}}(\mathbf{g}) = \text{clip}_{[0,1]} \left[\sum_{k=1}^K \alpha_k e^{-\frac{1}{2}((\Delta\theta - \mu_k)/\sigma_k)^2} C(\Delta\theta) \right], \quad (1)$$

where $C(\Delta\theta) = \cos^2\left(\frac{\Delta\theta}{\theta_{\text{fov}}/2} \frac{\pi}{2}\right)$ downweights peripheral detections. This converts fine-grained image evidence into an inspectable spatial distribution.

Two curiosity cues regulate exploration. Distance weighting $S_{\text{dist}}(\mathbf{g}) = 1/(1 + d(\mathbf{x}_r, \mathbf{g}))$ discourages unnecessary backtracking. Unknown-area weighting $S_{\text{unknown}}(\mathbf{g}) = 1 - e^{-k\rho_u(\mathbf{g})}$ favors candidates surrounded by unmapped reachable cells. Frontier goals are ranked by

$$S_{\text{NeSy}}(\mathbf{g}) = w_{\text{dist}} S_{\text{dist}}(\mathbf{g}) + w_{\text{VL}} S_{\text{VL}}(\mathbf{g}) S_{\text{unknown}}(\mathbf{g}). \quad (2)$$

Valid IBTPs are selected first by S_{VL} because their purpose is close-range verification; otherwise, the highest-scoring frontier is selected. FAR Planner [28] generates a collision-free route and replans as new obstacles appear.

IV. EXPERIMENTS

A. Experimental Setup

DARPA TIAMAT simulation. We test four scenarios on two simulators: Apartment 1 and Apartment 2 in Habitat-

TABLE I

PER-ENVIRONMENT SIMULATION AND PHYSICAL-ROBOT RESULTS. SR AND SPL ARE HIGHER-IS-BETTER; MTUR IS LOWER-IS-BETTER. DASHES DENOTE METHODS NOT DEPLOYED ON THE EDGE PLATFORM.

(a) DARPA TIAMAT simulation								
Method	Apartment 1		Apartment 2		Camping Site		Factory	
	SR (%)	MTUR	SR (%)	MTUR	SR (%)	MTUR	SR (%)	MTUR
Frontier	8.3	0.958	8.3	0.884	0.0	1.000	0.0	1.000
VLFM	8.3	0.931	8.3	0.864	4.2	0.953	8.3	0.859
SG-Nav	0.0	1.000	4.2	0.973	0.0	1.000	8.3	0.901
ApexNav	25.0	0.817	25.0	0.795	20.8	0.828	12.5	0.861
VL-Nav w/o IBTP	70.8	0.680	62.5	0.724	62.5	0.731	58.3	0.762
VL-Nav w/o curiosity	79.1	0.612	75.0	0.635	58.3	0.793	66.7	0.735
VL-Nav	87.5	0.562	79.2	0.591	75.0	0.647	75.0	0.679

(b) Physical-robot experiments								
Method	Hallway		Office		Apartment		Outdoor	
	SR (%)	SPL	SR (%)	SPL	SR (%)	SPL	SR (%)	SPL
Frontier	40.0	0.239	41.7	0.317	55.6	0.363	33.3	0.189
VLFM	53.3	0.366	75.0	0.556	66.7	0.412	44.4	0.308
VL-Nav w/o IBTP	66.7	0.593	83.3	0.738	70.2	0.615	55.6	0.573
VL-Nav w/o curiosity	73.3	0.612	86.3	0.743	66.7	0.631	55.6	0.498
VL-Nav	86.7	0.672	91.7	0.812	88.9	0.733	77.8	0.637

Sim [29], and a Camping Site and Factory in IsaacSim [30]. Each environment contains eight abstract navigation tasks, each evaluated for three trials. Instructions cover implicit semantics (rain $\rightarrow$ waterproof equipment), hierarchical search (garage $\rightarrow$ tools), and social concepts (black-tie party $\rightarrow$ suit, shoes, and tie), with some tasks requiring several targets at widely separated locations. A Boston Dynamics Spot with five RGB-D cameras provides surround perception. We report SR and max-time usage ratio (MTUR), the average execution time divided by the task limit.

Physical environments. We evaluate nine uncommon language-described targets over three to five trials in a hallway, office, apartment, and outdoor area. The main platform is a wheeled rover; a Unitree Go2 is used for the multi-floor demonstration. Both carry a Livox Mid-360 LiDAR and Intel RealSense RGB-D camera. The navigation stack runs on a Jetson Orin NX, while the NeSy Task Planner runs asynchronously on a remote device hosting Qwen3-VL-8B. We report SR and success weighted by path length (SPL).

Baselines are closest-frontier exploration, VLFM, SG-Nav, and ApexNav. We also test VL-Nav without IBTP and without curiosity. SG-Nav and ApexNav cannot run on the edge device and are therefore omitted from physical trials. For a fair real-time comparison, VLFM uses the same YOLO-World and FastSAM perception modules as VL-Nav rather than its original, heavier BLIP-2 model [31].

B. Results

Simulation. As shown in Table I, VL-Nav obtains 75.0–87.5% SR across the four scenarios, averaging 83.4% indoors, 75.0% outdoors, and 79.2% overall. The strongest baseline reaches at most 25.0% SR. SG-Nav has MTUR near 1.0 because repeated LLM chain-of-thought calls add latency; ApexNav lacks persistent symbolic memory for long-horizon, multi-target goals. The qualitative examples in Fig. 3 demonstrate that VL-Nav’s gain extends beyond direct category matching to implicit inference, hierarchical search, and abstract concept grounding.

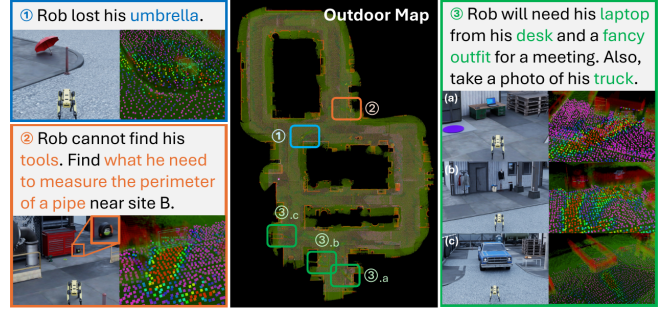


Fig. 4. Outdoor TIAMAT factory results. VL-Nav resolves direct retrieval, implicit inference (pipe perimeter $\rightarrow$ measuring tape), and a multi-object task whose targets occupy distant map regions. Colored point clouds visualize the semantic value distribution used for exploration.

Real world. VL-Nav achieves 86.3% mean SR and 0.714 mean SPL, including 91.7% SR and 0.812 SPL in the office. The consistent improvement over VLFM and frontier exploration indicates both reliable target discovery and less redundant travel. The system also completes a 483-m run and the multi-floor task in Fig. 1, which requires stairs, an object-reference relation, and recipient verification.

Ablations. IBTP is most important in cluttered environments: removing it lowers Apartment 1 simulation SR from 87.5% to 70.8% and physical-apartment SR from 88.9% to 70.2%. Without these semantic shortcuts, tentative targets are less likely to be approached and verified. Curiosity is most important in large environments: removing distance and unknown-area weighting lowers Factory SR from 75.0% to 66.7% and physical-outdoor SR from 77.8% to 55.6%. The two components therefore address complementary failure modes—semantic ambiguity near objects and inefficient wandering at map scale.

C. Qualitative Behavior and Generalization

The indoor examples in Fig. 3 expose three distinct uses of the unified memory. For the weather instruction, neural reasoning infers “rain jacket” before the symbolic graph supplies localized candidates. For the garage instruction, a room

node first provides a hierarchical intermediate goal; after reaching it, newly observed tool nodes support subsequent subtasks. For the black-tie instruction, the planner expands an abstract social concept into a formal suit, leather shoes, and tie, then retains completed targets while replanning for the remaining objects. These behaviors would be difficult to represent as a single fixed object-goal query.

The outdoor cases in Fig. 4 test a different regime: sparse visual evidence, long distances, and unstructured free space. VL-Nav first uses large, task-related landmarks to focus exploration and then approaches smaller IBTPs for verification. In the pipe task, the instruction names a measurement rather than an object, so the planner must infer that a measuring tape is required. In the multi-object task, targets occur in separated areas; the scene graph records completed discoveries, while the distance and unknown-area terms keep the robot from repeatedly crossing the explored map. The 75.0% SR and lower MTUR in both outdoor simulations indicate that the same policy remains effective when geometric scale and visual structure differ substantially from apartments.

Physical trials further test sim-to-real generalization across semantic complexity and scale: a cluttered office and apartment, a long hallway, and an unstructured outdoor area. The full system leads every environment in both SR and SPL. Its 483-m trajectory demonstrates that curiosity terms continue to discourage redundant long-range travel. The multi-floor run in Fig. 1 adds vertical transition, relational grounding, and person verification in one mission. Together, these trials show that the modular representation can absorb changes in robot platform, appearance, and environment without retraining the navigation policy.

D. Runtime and Interpretability

VL-Nav separates components by their required update rate. State estimation, mapping, open-vocabulary perception, goal selection, and collision avoidance run on the edge platform. The heavier VLM is queried only for initial decomposition, target verification, or replanning after a subtask completes. This asynchronous design avoids the latency accumulated by approaches that use chain-of-thought reasoning at every navigation step. It also makes the decision process inspectable: a failure can be localized to the inferred target list, retrieved scene-graph nodes, best-view images, semantic value field, selected goal, or geometric path, rather than appearing only as an unexplained action from an end-to-end policy.

E. Performance Factors

Environment scale. The physical environments span a small apartment, a medium office, and two large spaces (hallway and outdoor). Frontier exploration degrades from 55.6% SR and 0.363 SPL in the apartment to 36.7% mean SR and 0.214 mean SPL across the large settings. VL-Nav retains 82.3% mean SR and 0.655 mean SPL in those large spaces, compared with 88.9% and 0.733 in the apartment. This smaller degradation reflects the scoring policy: semantic

relevance focuses the search, unknown-area gain rewards informative views, distance discourages costly traversal, and IBTPs exploit promising objects before geometric coverage is complete.

Semantic structure. VL-Nav’s mean SR/SPL rises from 77.8%/0.637 in the low-context outdoor setting to 90.3%/0.773 across the semantically rich office and apartment. Structured rooms provide more graph relations, contextual cues for VLM verification, and informative open-vocabulary detections. Thus clutter, when encoded as explicit semantic memory, can become evidence for disambiguating language and selecting goals rather than only an obstacle to navigation.

Dynamic and edge operation. The scene graph represents longer-lived semantic task state, while LiDAR and terrain updates continuously revise the occupancy map and FAR Planner repairs routes invalidated by new obstacles. Progress monitoring triggers replanning when the robot is stuck. Meanwhile, the VLM stays off the critical control loop, observations are compressed into graph attributes and best-view images instead of an unbounded video history, and nearby IBTPs are voxel-downsampled before scoring. High-rate computation therefore remains limited to perception, mapping, and local planning, with expensive semantic de-liberation reserved for meaningful task boundaries. In our tests, standard 256×320 YOLO-World inference without TensorRT yielded about 23% higher SR than the tested higher-resolution FP16 modes, indicating that moderate resolution better balances recall and latency.

V. DISCUSSION AND CONCLUSION

VL-Nav offers a practical division of labor between semantic maps and foundation models. A large model performs occasional abstract reasoning; explicit scene structure preserves grounded history; and lightweight open-vocabulary perception continuously shapes exploration. The intermediate subtasks, graph queries, target nodes, and metric goals remain inspectable, which is valuable for diagnosing failures in long-horizon navigation.

The current system nevertheless has important limitations. It can struggle with descriptions that contain hidden spatial references or objects identified primarily by textual markings. The planner can only reason over evidence retained by the detector and best-view memory, so missed or poor-quality observations propagate to later decisions. Detection thresholds, lighting conditions, and scoring weights are hand tuned and may require recalibration in new sensing conditions. Finally, the graph assumes mostly persistent rooms and objects and does not yet model temporal relations or moving targets. Future work will incorporate temporal scene-graph updates, adaptive threshold and confidence calibration, and explicit uncertainty propagation from neural verification through symbolic planning and goal selection.

REFERENCES

- [1] E. Noorani, Z. Serlin, B. Price, and A. Velasquez, “From abstraction to reality: Darpa’s vision for robust sim-to-real autonomy,” *AI Magazine*, vol. 46, no. 2, p. e70015, 2025.
- [2] J. Gu, E. Stefani, Q. Wu, J. Thomason, and X. Wang, “Vision-and-language navigation: A survey of tasks, methods, and future directions,” in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2022, pp. 7606–7623.
- [3] A. Elfes, “Using occupancy grids for mobile robot perception and navigation,” *Computer*, vol. 22, no. 6, pp. 46–57, 1989.
- [4] C. Fennema and A. R. Hanson, “Experiments in autonomous navigation,” in *[1990] Proceedings. 10th International Conference on Pattern Recognition*, vol. 1. IEEE, 1990, pp. 24–31.
- [5] B. Yamauchi, “A frontier-based approach for autonomous exploration,” in *Proceedings 1997 IEEE International Symposium on Computational Intelligence in Robotics and Automation CIRA’97.*, 1997, pp. 146–151.
- [6] P. Mirowski, R. Pascanu, F. Viola, H. Soyer, A. J. Ballard, A. Banino *et al.*, “Learning to navigate in complex environments,” *arXiv preprint arXiv:1611.03673*, 2016.
- [7] F. Codevilla, M. Müller, A. López, V. Koltun, and A. Dosovitskiy, “End-to-end driving via conditional imitation learning,” in *2018 IEEE international conference on robotics and automation (ICRA)*. IEEE, 2018, pp. 4693–4700.
- [8] J. Zhang, K. Wang, S. Wang, M. Li, H. Liu, S. Wei, Z. Wang, Z. Zhang, and H. Wang, “Uni-navid: A video-based vision-language-action model for unifying embodied navigation tasks,” *arXiv preprint arXiv:2412.06224*, 2024.
- [9] A.-C. Cheng, Y. Ji, Z. Yang, X. Zou, J. Kautz, E. Biyik, H. Yin, S. Liu, and X. Wang, “Navila: Legged robot vision-language-action model for navigation,” in *RSS*, 2025.
- [10] J. Zhang, A. Li, Y. Qi, M. Li, J. Liu, S. Wang, H. Liu, G. Zhou, Y. Wu, X. Li *et al.*, “Embodied navigation foundation model,” *arXiv preprint arXiv:2509.12129*, 2025.
- [11] T. Gervet, S. Chintala, D. Batra, J. Malik, and D. S. Chaplot, “Navigating to objects in the real world,” *Science Robotics*, vol. 8, no. 79, p. eadf6991, 2023.
- [12] N. Yokoyama, S. Ha, D. Batra, J. Wang, and B. Bucher, “Vlfn: Vision-language frontier maps for zero-shot semantic navigation,” in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 42–48.
- [13] H. Yin, X. Xu, Z. Wu, J. Zhou, and J. Lu, “SG-Nav: Online 3D scene graph prompting for LLM-based zero-shot object navigation,” in *NeurIPS*, vol. 37, 2025, pp. 5285–5307.
- [14] M. Zhang, Y. Du, C. Wu, J. Zhou, Z. Qi, J. Ma, and B. Zhou, “Apex-nav: An adaptive exploration strategy for zero-shot object navigation with target-centric semantic fusion,” *IEEE Robotics and Automation Letters*, 2025.
- [15] S. Thrun, D. Fox, W. Burgard, and F. Dellaert, “Robust monte carlo localization for mobile robots,” *Artificial intelligence*, vol. 128, no. 1–2, pp. 99–141, 2001.
- [16] R. A. Newcombe, S. Izadi, O. Hilliges, D. Molyneaux, D. Kim, A. J. Davison, P. Kohi, J. Shotton, S. Hodges, and A. Fitzgibbon, “Kinectfusion: Real-time dense surface mapping and tracking,” in *2011 10th IEEE international symposium on mixed and augmented reality*. Ieee, 2011, pp. 127–136.
- [17] A. J. Davison, I. D. Reid, N. D. Molton, and O. Stasse, “Monoslam: Real-time single camera slam,” *IEEE transactions on pattern analysis and machine intelligence*, vol. 29, no. 6, pp. 1052–1067, 2007.
- [18] L. Freda and G. Oriolo, “Frontier-based probabilistic strategies for sensor-based exploration,” in *Proceedings of the 2005 IEEE International Conference on Robotics and Automation*. IEEE, 2005, pp. 3881–3887.
- [19] M. Savva, A. X. Chang, A. Dosovitskiy, T. Funkhouser, and V. Koltun, “Minos: Multimodal indoor simulator for navigation in complex environments,” *arXiv preprint arXiv:1712.03931*, 2017.
- [20] R. Ramrakhya, E. Undersander, D. Batra, and A. Das, “Habitat-web: Learning embodied object-search strategies from human demonstrations at scale,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 5173–5183.
- [21] D. S. Chaplot, D. P. Gandhi, A. Gupta, and R. R. Salakhutdinov, “Object goal navigation using goal-oriented semantic exploration,” *Advances in Neural Information Processing Systems*, vol. 33, pp. 4247–4258, 2020.
- [22] S. K. Ramakrishnan, D. S. Chaplot, Z. Al-Halah, J. Malik, and K. Grauman, “Poni: Potential functions for objectgoal navigation with interaction-free learning,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 18 890–18 900.
- [23] M. Hahn, D. S. Chaplot, S. Tulsiani, M. Mukadam, J. M. Rehg, and A. Gupta, “No RL, No Simulation: Learning to navigate without navigating,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021, pp. 26 661–26 673.
- [24] N. Hughes, Y. Chang, S. Hu, R. Talak, R. Abdulhai, J. Strader, and L. Carlone, “Foundations of spatial perception for robotics: Hierarchical representations and real-time systems,” *The International Journal of Robotics Research*, vol. 43, no. 10, pp. 1457–1505, 2024.
- [25] S. Bai, Y. Cai, R. Chen, K. Chen, X. Chen, Z. Cheng, L. Deng, W. Ding, C. Gao, C. Ge *et al.*, “Qwen3-vl technical report,” *arXiv preprint arXiv:2511.21631*, 2025.
- [26] T. Cheng, L. Song, Y. Ge, W. Liu, X. Wang, and Y. Shan, “Yolo-world: Real-time open-vocabulary object detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 16 901–16 911.
- [27] X. Zhao, W. Ding, Y. An, Y. Du, T. Yu, M. Li, M. Tang, and J. Wang, “Fast segment anything,” *arXiv preprint arXiv:2306.12156*, 2023.
- [28] F. Yang, C. Cao, H. Zhu, J. Oh, and J. Zhang, “Far planner: Fast, attemptable route planner using dynamic visibility update,” in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2022, pp. 9–16.
- [29] M. Savva, A. Kadian, O. Maksymets, Y. Zhao, E. Wijmans, B. Jain, J. Straub, J. Liu, V. Koltun, J. Malik *et al.*, “Habitat: A platform for embodied ai research,” in *Proceedings of the IEEE/CVF international conference on computer vision*, 2019, pp. 9339–9347.
- [30] NVIDIA, “Isaac sim,” <https://github.com/isaac-sim/IsaacSim>, 2025.
- [31] J. Li, D. Li, S. Savarese, and S. Hoi, “Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models,” in *International conference on machine learning*. PMLR, 2023, pp. 19 730–19 742.